

Project 07 — Find Task-Sensitive Channels on a Sensor Grid

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given 200 trials of responses from an 8x8 sensor grid (64 channels). Each trial has a task condition. Your job is to find which channels respond to the task, and visualise the result on the 8x8 grid.

Background (short)

For each channel independently, we fit a simple linear model:

$$\text{response} = \text{beta}_0 + \text{beta}_1 * \text{condition} + \text{noise}$$

If beta_1 is far from zero for a channel, that channel responds to the task. We measure this with a t-statistic.

Data

In `shared/data/`:

- `trials.csv` — `trial_index`, `block_index`, `condition`, `trial_role`.
- `channel_responses.csv` — `trial_index` and 64 channel response columns (`ch00...ch63`).
- `channel_positions.csv` — the (row, col) position of each channel on the 8x8 grid.

Steps

1. Load and look. Open the starter notebook. Load all three files. Print their shapes.
2. Fit a GLM per channel. For each of the 64 channels, fit $\text{response} = \text{beta}_0 + \text{beta}_1 * \text{condition}$ using `scipy.stats.linregress` or `statsmodels`. Save beta_1 and its t-statistic.
3. Grid the t-statistics. Reshape the 64 t-stats into an 8x8 matrix using the channel positions. Plot as a heatmap.
4. Threshold and count. Count how many channels have $|t| > 2$. Mark them on the heatmap.

What to hand in

- Your filled notebook.
- The 8x8 heatmap of t-statistics.

- A count of the significant channels.
- A short paragraph (3-4 sentences) on where in the grid the response is located.

If you finish early: apply a Bonferroni correction (threshold $|t|$ where $p < 0.05 / 64$) and see if any channels survive.